

Meshkat Shariat Bagheri

AI Engineer | Agentic AI | Local LLMs | RAG | NLP | OCR

Tehran, Iran | +98 902 111 9991 | meshkatshb@gmail.com

[linkedin.com/in/meshkatshb](https://www.linkedin.com/in/meshkatshb) | github.com/meshkatshb | meshkatshb.github.io | [Google Scholar](#)

PROFESSIONAL SUMMARY

AI Engineer focused on enterprise-grade **Agentic AI**, **local LLM systems**, and **RAG applications**. Experienced in building production-ready local-first AI agents integrated with OracleDB, Microsoft CRM/Dynamics 365, RocketChat, Zulip, Exchange/EWS, and internal document systems. Delivered agents for protected document search, CRM case retrieval and analytics, quality scoring, document review, natural-language-to-database search, invoice generation, meeting-minutes generation, page-aware browser workflows, OCR pipelines processing **250k+ records**, and Persian ASR. Strong technical background in Python, LangChain, LangGraph, Ollama, Llama.cpp, PyTorch, Transformers, Milvus hybrid search, and GNN-based NLP research.

TECHNICAL SKILLS

- **AI / LLM:** Agentic AI, LLMs, RAG, AI Agents, Tool Calling, Prompt Engineering, LLM Evaluation, LLMOps, Local LLM Deployment, Fine-tuning
- **NLP / ML:** NLP, OCR, ASR, Information Retrieval, Semantic Search, Embeddings, Vector Search, Text Summarization, GNNs, Deep Learning, Machine Learning
- **Frameworks:** LangChain, LangGraph, vLLM, PyTorch, Transformers, Ollama, Llama.cpp, Streamlit, Gradio, FastAPI, Django REST Framework
- **Databases:** FAISS, Milvus, Qdrant, OracleDB, PostgreSQL, MySQL, Redis
- **Language and Tools:** Python, SQL, JavaScript, TypeScript, Go, Git, Docker, Linux, Nginx, Swagger, Jira

PROFESSIONAL EXPERIENCE

AI Engineer

Jun 2025 – Present

Gam Electronics – Almas Team

Tehran, Iran

- Built a fully local enterprise RAG agent platform using reasoning models, company documents, MCP/A2A-style tool catalogs, agent cards, manual LangChain tool-calling loops, streamed responses, conversation memory, and tool-call traces.
- Integrated AI agents with organizational messaging systems (**RocketChat** and **Zulip**), Microsoft CRM/Dynamics 365, Exchange/EWS, and Oracle-backed internal document systems for support and case workflows.
- Engineered an access-controlled letter-search agent with Exchange login, Oracle email-to-user access mapping, per-letter permission checks, full-text reconstruction from Milvus chunks, source previews, and feedback analytics.
- Developed an OCR pipeline using DeepSeek-OCR and other OCR/VLM models to extract text from 250k+ OracleDB document records, each containing multiple scanned pages.
- Improved Milvus retrieval quality with dense + BM25 hybrid search, multi-concept AND ranking, adaptive score thresholds, Persian exact-match boosts, duplicate-chunk cleanup, indexed-letter discovery, and nightly incremental indexing jobs.
- Designed an agentic document-review and fact-checking system that ingests PDF, Word, PowerPoint, HTML, TXT, Markdown, and CSV files; extracts claims; stores Qwen3 embeddings in Milvus; and ranks contradiction, missing-evidence, and logic-change findings.
- Built a natural-language-to-OracleDB search application that maps user prompts to database fields, improving user satisfaction by nearly 30% through Qwen3:8B fine-tuning and enhanced LangExtract-style extraction.
- Implemented CRM resolved-case agents and analytics data products for semantic case retrieval, full case-detail loading, relative date filters, quality scoring, expert feedback, aggregate statistics, monthly trends, customer/unit/status breakdowns, topic clustering, and Shamsi metadata repair.
- Developed productivity agents for invoice generation, meeting-minutes generation from audio, page-aware Chrome workflows, task import, follow-up email generation, stand-up summaries, and email RAG over indexed inbox content.
- Automated production maintenance, analytics, and observability workflows including OCR/no-OCR index jobs, lookback windows for delayed Oracle rows, dynamic FAQ generation from usage logs via RemoteEmbeddings + Louvain/greedy clustering, local Langfuse tracing, scoring observations, and shared evaluation hooks.
- Implemented a Persian ASR application using **VOSK** to convert voice input into structured transcriptions.
- **Stack:** Python, LangChain, LangGraph, Ollama, Llama.cpp, PyTorch, Transformers, FastAPI, Flask, Streamlit, Gradio, React, Milvus, FAISS, SQLite, OracleDB, Dynamics 365, OCR, RAG, LLM fine-tuning.

Software Engineer

Jun 2023 – Oct 2023

Gam Electronics – Zomorrod Team

Tehran, Iran

- Boosted system performance by 30% and responsiveness by 15% via architecture and feature enhancements.
- **Stack:** .NET, Jira, Swagger, Git

Software Developer

Aug 2019 – Mar 2020

Najva – Retention Marketing Platform

Tehran, Iran

- Improved backend service speed by up to 30% through Python/Django optimization and refactoring.
- Reduced report query time by 25%, improving backend efficiency and user experience.
- Improved code quality by 40% by applying clean-code practices, PEP8, and collaborative review workflows.
- **Stack:** Python, Django REST Framework, Linux, Nginx, Redis, Git, Jira.

PROJECTS

remote-embedding (GitHub, PyPI) | Python, FastAPI, LangChain, Hugging Face, PyPI Apr 2026 – Present

- Published a Python package that exposes a shared FastAPI embedding service and a LangChain-compatible RemoteEmbeddings client for RAG applications.
- Enabled multiple applications to reuse one server-side Hugging Face embedding model instance, reducing duplicated VRAM usage on limited GPU hardware.
- Implemented configurable CLI/environment options for model selection, local model cache paths, CPU/GPU device choice, batching, request limits, CUDA cache cleanup, model kwargs, and encode kwargs.

Open-Source Contributor – Google LangExtract (GitHub) | Python, Git Aug 2025 – Present

- Added Llama.cpp support to enable local CPU-based model execution.
- Improved multilingual rendering support for Persian, Chinese, and Japanese.
- Contributed to local-first information extraction workflows for LLM-based applications.

Open-Source Contributor – browser-use WebUI (GitHub) | Python, Git Jan 2025 – Present

- Improved UX for a 55k+ star open-source browser-agent project by adding default LLM configurations and suppressing unnecessary warnings.
- Refactored LLM configuration handling to improve maintainability and reduce setup friction.
- Fixed runtime issues, including UnboundLocalError cases affecting browser startup and shutdown.
- Updated CLI documentation for theme-related options.

AI / LLM Learning Repository – Zero-to-Mastery Series (GitHub) | Python, NLP, ML, Git Jul 2024 – Present

- Created a hands-on learning series covering Python, LLMs, NLP, and software development.
- Includes real-world projects and structured tutorials. Featured: LangChain, LLM.

EDUCATION

University of Tehran

Tehran, Iran

M.Sc. in Information Technology Engineering

Sep 2020 - Sep 2024

- *Thesis Title:* "Text Summarization with Graph Neural Networks;" Introduced a novel Community Detection Long-Text Summarization (CDLTS) model combining community detection techniques with GAT.
- *Supervisor:* Dr. Salehi
- *Relevant Coursework:* Text Mining(A+), Machine Learning(A), Social Networks(A+), Complex Networks(A), Information Network Analysis (A)
- *GPA w/o failed courses:* 3.61 (Last two semesters: 3.75)

University of Tehran

Tehran, Iran

B.Sc. in Computer Engineering

Sep 2015 - Sep 2020

- *Thesis Title:* Implementation of the OpenStack Tool as an Internet of Things (IoT) Platform
- *Supervisor:* Dr. Khonsari
- *Relevant Coursework:* AI, Internet Engineering, Data Structure, Algorithm Design, Advanced Programming

PUBLICATIONS

DCOR: Anomaly Detection in Attributed Networks via Dual Contrastive Learning Reconstruction, H. RafieiZade, H. Zare, M. Ghassemi Parsa, H. Davardoust, and M. Shariat Bagheri. Complex Networks & Their Applications XIII. COMPLEX NETWORKS 2024 2024. Studies in Computational Intelligence, vol 1189. Springer, Cham. Available at [Springer Chapter](#).